MTHFD1 Deficiency — Comprehensive Disease Characteristics Report

Disease: MTHFD1 Deficiency (Methylenetetrahydrofolate Dehydrogenase 1 Deficiency) **Category:** Mendelian (autosomal recessive inborn error of folate metabolism / inborn error of immunity) **Report type:** Literature-based knowledge synthesis (no primary patient data provided) **Date:** 2026-09-04

> **Evidence-source note:** Because this is an ultra-rare disorder (<20 molecularly confirmed patients worldwide as of the primary literature reviewed), most clinical statements derive from individual case reports and small case series (human clinical evidence). Mechanistic statements draw on patient-fibroblast studies (in vitro) and mouse models (model organism). Where a claim is inferred rather than demonstrated, this is stated explicitly.

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1. Disease Information

Overview. MTHFD1 deficiency is a rare autosomal recessive inborn error of the **cytoplasmic (and nuclear) folate cycle** caused by biallelic loss-of-function variants in *MTHFD1*. The gene encodes a **trifunctional enzyme** carrying three catalytic activities: 5,10-methylenetetrahydrofolate dehydrogenase, 5,10-methenyltetrahydrofolate cyclohydrolase, and 10-formyltetrahydrofolate synthetase. Loss of function disrupts the supply of one-carbon folate coenzymes required for **de novo purine synthesis, de novo thymidylate (dTMP) synthesis, and remethylation of homocysteine to methionine**. Clinically it manifests as a multisystem, folate-responsive disorder combining **megaloblastic anemia, combined/severe combined immunodeficiency, atypical hemolytic-uremic syndrome (aHUS), hyperhomocysteinemia, and neurologic abnormalities** (P 21813566; 25548164; 32414565).

It was first described in 2011 by Watkins et al., who identified it via exome sequencing in a single infant — "the first case of an inborn error of folate metabolism affecting the trifunctional MTHFD1 protein" (P 21813566).

Key identifiers. - **Mondo: MONDO:0060611** — "combined immunodeficiency and megaloblastic anemia with or without hyperhomocysteinemia" (acronym **CIMAH**) *(verified via EBI OLS4/MONDO)* - **OMIM (phenotype): #617780** — same name - **OMIM (gene):** *MTHFD1* **172460** - **Orphanet: ORPHA:658813** - **UMLS:** C4540434 · **GARD:** 0026001 · **MedGen:** 1615364 - **HGNC:** 7432 · **NCBI Gene:** 4522 · **Ensembl:** ENSG00000100714 · **UniProt:** P11586 · **cytoband:** 14q23.3 *(verified via MyGene.info)* - **ICD-11:** 4A00.xx (combined immunodeficiencies) / 5C50.x (metabolic disorders) — no dedicated code; **ICD-10:** D81.x (combined immunodeficiencies) / E53.8 as closest approximations - **MeSH:** no dedicated descriptor; indexed under "Immunologic Deficiency Syndromes," "Anemia, Megaloblastic," and "Folic Acid" metabolism terms

Synonyms / alternative names. MTHFD1 deficiency; Methylenetetrahydrofolate dehydrogenase 1 deficiency; **CIMAH** (Combined ImmunodeficiencY and Megaloblastic Anemia with or without Hyperhomocysteinemia); MTHFD1-related folate metabolism disorder. Gene aliases: MTHFC, MTHFD.

Data source type. Aggregated disease-level knowledge derived from published individual patient reports and biochemical/model studies — not EHR-derived population data.

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2. Etiology

Primary cause (genetic). Biallelic (homozygous or compound heterozygous) pathogenic variants in *MTHFD1* (14q23.3). Reported causal variants include missense, nonsense, and splice-site changes distributed across the dehydrogenase/cyclohydrolase and synthetase domains (see §4). The defect causes markedly reduced MTHFD1 protein and **absent methylenetetrahydrofolate dehydrogenase activity** in patient cells (P 32414565).

Genetic risk factors. - *Causal:* rare biallelic loss-of-function *MTHFD1* variants (the disease itself). - *Common susceptibility (distinct entity):* **MTHFD1 R653Q (c.1958G>A, rs2236225)** in the synthetase domain — a low-penetrance hypomorphic variant, homozygous in ~20% of individuals of European ancestry, associated with neural tube defects (NTDs), congenital heart defects (CHDs), and adverse pregnancy outcomes (P 23704330 ; 26408344). It acts predominantly as a **maternal** risk factor: maternal QQ homozygosity is associated with NTD-affected pregnancy at **OR ~1.5** (original Irish cohort OR 1.52, 95% CI 1.16–1.99, P=0.003, P 12384833 ; independent replication OR 1.49, 95% CI 1.07–2.09, P=0.019, P 16552426 ; meta-analysis of 9 studies / 4,302 cases / 4,238 controls confirming a maternal-allele excess, P 24977710). A promoter SNP (rs1076991) further modifies risk in combination with R653Q (P 19130090). This is a **modifier/susceptibility allele, not the Mendelian disease**. - *Consanguinity:* homozygous cases have been reported in consanguineous/founder settings (e.g., Kuwaiti children homozygous for c.517C>T p.Arg173Cys; P 42301236), consistent with AR disease.

Environmental risk factors. Maternal/dietary **folate deficiency** can exacerbate one-carbon metabolic insufficiency; in mouse models maternal folate-deficient diet interacts with synthetase deficiency (P 26408344). **Arsenic** (arsenic trioxide) directly targets MTHFD1/SUMO-dependent nuclear dTMP biosynthesis, a teratogenic mechanism converging on the same pathway (P 28265077).

Protective factors. Folate/folinic acid supplementation is the principal disease-modifying (protective) factor — it partially restores one-carbon flux and improves hematologic and immune outcomes (P 23296427 ;

42301236). Adequate maternal folate status is protective against the R653Q-associated developmental risks (inferred from GxE data, P 26408344).

Gene–environment interaction. Folate availability modulates the phenotype at every level: for the R653Q hypomorph, folate deficiency increases risk of developmental defects; for the Mendelian disease, exogenous reduced folate (folinic acid) partially bypasses the block. Arsenic × folate is a documented toxicant–pathway interaction (P 28265077).

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3. Phenotypes

Onset is typically **neonatal to infantile**; severity and progression are **variable** (from lethal early infancy to milder, treatment-responsive disease). HPO term suggestions and approximate frequencies (from the small published cohort) below.

Phenotype	Type	Onset / course	Approx. frequency	HPO suggestion
Megaloblastic anemia	Lab / hematologic	Infantile; responsive to therapy	Near-universal (hallmark)	HP:0001889
Combined / severe combined immunodeficiency	Clinical / immune	Infantile; progressive if untreated	Very common	HP:0005387 / HP:0004430
Recurrent infections (bacterial, sinopulmonary; opportunistic incl. *Pneumocystis*)	Clinical sign	Infantile	Common	HP:0002719
Lymphopenia (all subsets), poor vaccine response	Lab	Infantile	Common	HP:0001888
Atypical hemolytic-uremic syndrome / thrombotic microangiopathy	Clinical / lab	Infancy–childhood, episodic	Subset (several patients)	HP:0005575 / HP:0001937
Hyperhomocysteinemia	Lab	Congenital-biochemical	Frequent (not universal)	HP:0002160
Hypomethioninemia	Lab	Biochemical	Reported	HP:0500152 (low methionine)
Neurologic abnormalities (seizures, developmental delay)	Clinical	Infantile	Subset	HP:0001250 / HP:0001263
Failure to thrive	Clinical	Infantile	Common	HP:0001508
Autoimmune disease (e.g., autoimmune thyroiditis)	Clinical	Childhood	Subset	HP:0002960
Retinopathy	Clinical	Variable	Rare (1 patient)	HP:0000488
Liver fibrosis		Childhood	Rare	HP:0001395

Phenotype	Type	Onset / course	Approx. frequency	HPO suggestion
	Clinical / path			
Severe metabolic acidosis	Lab	Neonatal (severe cases)	Rare (lethal cases)	HP:0002153

Evidence:

P 21813566 ;
23296427; 25633902; 32414565; 42301236.

Quality-of-life impact. Untreated disease is life-threatening (recurrent/opportunistic infection, bone-marrow failure, TMA/renal injury, neurodevelopmental impairment). With early folate/folinic acid therapy, hematologic and immune function can substantially normalize, allowing discontinuation of anti-infective prophylaxis (P 27707659). No formal EQ-5D/SF-36 data exist for this ultra-rare disease.

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4. Genetic / Molecular Information

Causal gene. *MTHFD1* (HGNC:7432; OMIM 172460; NCBI Gene 4522; Ensembl ENSG00000100714), chromosome **14q23.3**, encoding the ~101 kDa trifunctional protein C1-THF synthase (UniProt P11586). N-terminal domain: dehydrogenase + cyclohydrolase (NADP-binding); C-terminal domain: 10-formyltetrahydrofolate synthetase.

Reported pathogenic variants (germline, biallelic). - c.517C>T, **p.Arg173Cys** (homozygous; Kuwaiti patients) —

P 42301236

- c.517C>T, **p.Arg173Cys** vicinity / **p.R173C** critical NADP-binding arginine + c.727+1G>A splice (first proband) —

P 21813566

- c.806C>T, **p.Thr296Ile** + c.1674G>A splice (exon skipping) —

P 25633902

- c.146C>T, **p.Ser49Phe** + c.673G>T, **p.Glu225*** (nonsense) —

P 25633902

- Compound heterozygous missense + exon-13 deletion (structural) —

P 27707659

- Additional deleterious compound-heterozygous variants —

P 32414565

Variant classes: missense, nonsense, canonical splice-site, and exonic deletions — all consistent with **loss of function** (no dominant-negative/gain-of-function reported). ACMG classification: reported variants are pathogenic/likely pathogenic based on segregation, functional enzymology, and predicted deleteriousness.

ClinVar landscape (reference transcript NM_005956.4; queried Iteration 4). 64 **Pathogenic** and 30 **Likely pathogenic** records (vs 629 VUS and 590 Benign among 681 total gene entries). The disease-associated P/LP spectrum is **overwhelmingly loss-of-function/null**: among 64 P/LP records — SNVs 39 (predominantly canonical splice ±1/±2, e.g., c.377+1G>C, c.616-2A>C, c.2280-1G>T, c.1264+1G>A; plus nonsense c.316G>T p.Glu106Ter, c.886G>T p.Glu296Ter), frameshift deletions 10 (e.g., c.2375del p.Gly792fs, c.1755_1756del p.Arg585fs, c.153_154del p.Ile53fs), duplications 2 (c.731dup p.Asp244fs; c.253dup p.Ile85fs), insertion 1 (c.2479_2480insTTGCACA p.Arg827fs), and 12 large copy-number gains/losses (chromosomal, largely incidental). Many P/LP entries are explicitly annotated to the trait "Combined immunodeficiency and megaloblastic anemia." The large VUS pool reflects the gene's tolerance of common missense variation (e.g., R653Q) and complicates novel-missense classification. *(Source: NCBI ClinVar via E-utilities.)*

Functional consequence. Loss of function: MTHFD1 protein reduced to **4.8–14.3% of control** (one patient ~44%) with **no detectable dehydrogenase activity** in fibroblasts (P 32414565).

Allele frequency. Rare disease alleles are absent/ultra-rare in gnomAD. In contrast, the common non-disease modifier **R653Q (rs2236225; GRCh38 14-64442127-G-A)** is very common — **gnomAD v4 global genome AF 0.386**, with ancestry-specific AF: **Non-Finnish European 0.453 (≈20.5% QQ homozygotes under Hardy-Weinberg)**, Admixed American 0.519, South Asian 0.505, Finnish 0.446, Ashkenazi Jewish 0.436, Middle Eastern 0.432, East Asian 0.221, African/African-American 0.212 *(verified via gnomAD v4 API; consistent with the "~20% of Caucasians homozygous" statement in*

P 23704330

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Somatic vs germline. Disease variants are **germline**. (Somatic relevance: MTHFD1 is exploited in cancer one-carbon metabolism and is targeted by arsenic trioxide —

P 28265077

— but this is not the inherited disease.)

Modifier genes / epigenetics / chromosomal abnormalities. No formally established modifier genes beyond folate-pathway context. Downstream **DNA hypomethylation** is expected from reduced methionine/SAM (mechanistically inferred; not systematically profiled in patients). No recurrent chromosomal abnormalities; one causal exon-13 deletion is a small intragenic structural variant (

P 27707659

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5. Environmental Information

- **Environmental factors:** dietary **folate deficiency** aggravates the metabolic block (GxE;

  P 26408344

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  Arsenic targets the same nuclear dTMP pathway (

  P 28265077

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- **Lifestyle factors:** maternal folate intake is the key modifiable factor for pathway-related developmental risk. Alcohol (a folate antagonist) and antifolate drugs would be expected to worsen one-carbon insufficiency (inferred).
- **Infectious agents:** none causal. Infections are **consequences** of the immunodeficiency (recurrent bacterial sinopulmonary infections; opportunistic organisms including *Pneumocystis jirovecii*, consistent with a SCID-like state).

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6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic loss-of-function *MTHFD1* variants** reduce MTHFD1 protein to ~5–14% of normal and abolish dehydrogenase activity (**demonstrated**, P 32414565).
2. This **impairs interconversion of one-carbon–substituted tetrahydrofolate coenzymes** (methylene-THF ↔ methenyl-THF ↔ 10-formyl-THF) in the cytosol/nucleus (**demonstrated** enzymology).
3. Reduced supply of **5,10-methylene-THF** and **10-formyl-THF** limits three downstream outputs; the pathway then **branches**:
4. **Branch A — Nuclear de novo thymidylate (dTMP) synthesis falls (~50% reduced flux) → uracil is misincorporated into DNA** → futile base-excision-repair cycles and genomic stress (**demonstrated**, P 25548164). → **leads to** ineffective erythropoiesis (**megaloblastic anemia**) and impaired lymphocyte proliferation (**combined/severe combined immunodeficiency**).
5. **Branch B — Homocysteine remethylation to methionine falls (~90% reduced formate→methionine flux) → hyperhomocysteinemia and low methionine/SAM (demonstrated**, P 25548164). → endothelial injury contributes to **thrombotic microangiopathy / atypical HUS (inferred** from homocysteine's known endothelial toxicity) and reduced methylation capacity contributes to **neurologic abnormalities (inferred**).
6. **Branch C — De novo purine synthesis is relatively spared** in patient fibroblasts (formate→purine flux unaffected), explaining why purine-dependent phenotypes are less prominent than dTMP-dependent ones (**demonstrated**, P 25548164).
7. The combined hematologic failure, immune failure, and vascular/renal injury → **recurrent/ opportunistic infection, bone-marrow failure, renal impairment, failure to thrive, and neurodevelopmental impairment (human clinical,**

P 23296427 ;
42301236).

8. **Intervention branch:** exogenous **reduced folate (folinic acid)** replenishes downstream one-carbon pools, partially restoring dTMP/methionine synthesis → **improved hematology and immune reconstitution** (**demonstrated**,
P 27707659 ;
23296427).

Category detail (checklist)

- **Molecular pathways:** folate/one-carbon metabolism (KEGG hsa00670 one-carbon pool by folate); methionine cycle; de novo purine (IMP) and pyrimidine (dTMP) biosynthesis. **Reactome:** metabolism of folate and pterines.
- **Cellular processes:** DNA replication/repair (uracil misincorporation, BER), cell-cycle arrest in rapidly dividing precursors (erythroid, lymphoid), impaired proliferation. GO suggestions: GO:0006730 (one-carbon metabolic process), GO:0046655 (folic acid metabolic process), GO:0006231 (dTMP biosynthetic process), GO:0009086 (methionine biosynthesis), GO:0006189 (de novo IMP biosynthesis).
- **Protein dysfunction:** loss of function / markedly reduced protein abundance and abolished dehydrogenase activity; MTHFD1 is normally partitioned between cytosol and nucleus (SUMO-dependent nuclear import for dTMP synthesis) (P 25548164 ; 28265077).
- **Metabolic changes:** ↓ 10-formyl-THF, ↓ methionine/SAM, ↑ homocysteine, ↑ uracil in DNA; elevated methylmalonic acid and selectively decreased methylcobalamin synthesis were noted in the index patient (P 21813566).
- **Immune system involvement:** combined immunodeficiency with lymphopenia across subsets and poor vaccine responses (P 27707659) plus paradoxical **autoimmunity** (thyroiditis, autoimmune disease) — dysregulation of both arms.

- **Tissue damage mechanisms:** endothelial/microvascular injury (TMA/aHUS); ineffective hematopoiesis; possible hepatic fibrosis.
- **Biochemical abnormality:** trifunctional folate enzyme deficiency (EC 1.5.1.5 / EC 3.5.4.9 / EC 6.3.4.3).
- **Epigenetic changes:** reduced SAM → expected global/DNA hypomethylation (**inferred**, not systematically profiled).
- **Molecular profiling / functional genomics:** patient-fibroblast flux studies (P 25548164); MTHFD1 as an arsenic and antifolate metabolic target and a dependency in some cancers (P 28265077).

Cell types (CL suggestions): erythroid progenitor (CL:0000038), T cell (CL:0000084), B cell (CL:0000236), hematopoietic stem/progenitor cell (CL:0000037), vascular endothelial cell (CL:0000115).

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7. Anatomical Structures Affected

- **Organ / system level (primary):** hematopoietic/immune system — **bone marrow** (UBERON:0002371), **blood** (UBERON:0000178). **Kidney** (UBERON:0002113) via aHUS/TMA. **Central nervous system / brain** (UBERON:0000955) via neurologic features.
- **Secondary involvement: retina/eye** (UBERON:0000970) — retinopathy; **liver** (UBERON:0002107) — fibrosis; **lungs** (recurrent pneumonia); **thyroid** (autoimmune thyroiditis).
- **Tissue/cell level:** erythroid and lymphoid lineages; vascular endothelium (microangiopathy).
- **Subcellular level (GO Cellular Component): cytosol** (GO:0005829) and **nucleus** (GO:0005634) — MTHFD1 functions in both compartments; nuclear pool drives de novo dTMP synthesis (P 25548164). Note: MTHFD1 is cytoplasmic/nuclear, distinct from the mitochondrial paralog MTHFD2/ MTHFD1L.

- **Localization / lateralization:** systemic and **bilateral** (hematologic, immune, metabolic); renal and retinal involvement generally bilateral.

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8. Temporal Development

- **Onset:** congenital/neonatal to infantile; **subacute-to-chronic** presentation with acute decompensations (infection, TMA, acidosis).
- **Progression:** without treatment, **progressive** with life-threatening bone-marrow failure and infection; severe neonatal cases can be **rapidly fatal** (siblings dead at 9 weeks with megaloblastic anemia, infection, severe acidosis; P 25633902; a 3-year-old died on day 18 of PICU care, P 42301236).
- **Course pattern:** chronic underlying metabolic defect with **episodic** TMA/aHUS crises; largely **treatment-responsive** when folate therapy is started early.
- **Critical period / window of opportunity: early molecular diagnosis and prompt folinic/folic acid initiation** is the key intervention window — determines survival and degree of immune reconstitution (P 42301236; 27707659).

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9. Inheritance and Population

- **Inheritance: autosomal recessive** (biallelic loss-of-function). Both parents obligate heterozygous carriers; unaffected sib carrying neither variant supports segregation (P 21813566).
- **Penetrance / expressivity:** biallelic LOF appears **highly penetrant** but with **variable expressivity** (severity ranges from lethal infancy to milder folate-responsive disease), even

within families (25633902).

- **Epidemiology: ultra-rare** — fewer than ~20 molecularly confirmed patients reported worldwide in the reviewed literature; true prevalence/incidence unknown (no registry estimates). No published incidence per 100,000. (For context, the *R653Q modifier*-associated **neural tube defects** occur in ~1 in 1000 pregnancies in the US/Europe, and maternal periconceptional folic acid reduces NTD occurrence by 50–70% — 22856873; but these are pathway-level, not MTHFD1-deficiency, figures.)
- **Founder / consanguinity:** homozygous cases reported in consanguineous families (e.g., Kuwaiti children, p.Arg173Cys; 42301236); no established broad founder allele.
- **Carrier frequency:** not established for pathogenic LOF alleles (ultra-rare). By contrast the common **R653Q** modifier reaches ~50% allele frequency in Europeans (23704330) — but this is **not** the disease allele.
- **Sex ratio / age distribution:** no sex bias reported; affected individuals are predominantly infants/young children.
- **Geographic distribution:** cases reported from Europe, North America, and the Middle East; no defined endemic region.
- **Variant-frequency geography (R653Q modifier, gnomAD v4):** the common R653Q Q-allele is most frequent in Admixed American (0.519), South Asian (0.505), Non-Finnish European (0.453), Finnish (0.446), Ashkenazi (0.436) and Middle Eastern (0.432) populations, and least frequent in East Asian (0.221) and African/African-American (0.212) populations — potentially relevant to population-specific folate-pathway developmental risk (verified via gnomAD v4).

10. Diagnostics

Laboratory / biochemical. - **CBC + blood smear:** macrocytic **megaloblastic anemia** (± pancytopenia), hypersegmented neutrophils (LOINC macrocyte/MCV panels). - **Plasma total homocysteine:** elevated (hyperhomocysteinemia) in many patients; **methionine** low/normal. - **Methylmalonic acid:** elevated in the index case; **methylcobalamin synthesis** decreased in cultured fibroblasts (P 21813566).
- **Immunology:** lymphopenia across T/B/NK subsets, hypogammaglobulinemia/poor vaccine responses (P 27707659).
- **Renal / hemolysis markers** during aHUS/TMA (schistocytes, ↑LDH, ↑creatinine, thrombocytopenia). - **Cellular biomarkers (research):** absent MTHFD1 dehydrogenase activity and reduced protein by Western blot in fibroblasts; **elevated uracil in DNA**; abnormal formate-incorporation flux assays (P 32414565; 25548164).

Genetic testing (definitive). Diagnosis is confirmed by identifying **biallelic *MTHFD1* variants**. Recommended approach: **whole-exome sequencing (WES)** or a **combined immunodeficiency / inborn-errors-of-metabolism gene panel** including *MTHFD1*; WGS or targeted analysis can detect **intragenic structural variants** (e.g., exon-13 deletion missed by standard SNV calling — requires read-depth/CNV analysis) (P 27707659; 21813566; 23296427). Single-gene sequencing of *MTHFD1* is appropriate when phenotype is characteristic. CMA/karyotype/FISH generally not informative (defect is intragenic). Mitochondrial DNA and repeat-expansion testing: not applicable.

Clinical criteria / differential diagnosis. No formal consensus criteria. Consider MTHFD1 deficiency in any infant with **megaloblastic anemia** + **immunodeficiency** + **hyperhomocysteinemia and/or aHUS**. Differentials: other inborn errors of folate/cobalamin metabolism (**hereditary folate malabsorption** [SLC46A1], **cblC/MMACHC** and related cobalamin defects, **transcobalamin deficiency**, **DHFR deficiency**, **MTHFR deficiency**), other SCID genotypes, and complement-mediated aHUS. Distinguishing features: MTHFD1 uniquely combines the folate-cycle biochemistry (low methionine, high Hcy, normal purine flux) with combined immunodeficiency (P 32412981).

Screening. Not part of standard newborn screening. Cascade carrier testing of relatives once the familial variants are known; prenatal/preimplantation testing feasible for known biallelic variants.

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11. Outcome / Prognosis

- **Survival / mortality:** highly variable and **diagnosis-timing–dependent**. Early-diagnosed, promptly treated patients can achieve immune reconstitution and good outcomes; delayed diagnosis carries high mortality (neonatal deaths and a PICU death reported) (P 25633902; 42301236).
- **Morbidity:** infections, bone-marrow failure, renal injury from TMA/aHUS, neurodevelopmental impairment, and rarely retinopathy/liver fibrosis.
- **Recovery potential:** hematologic and immune parameters are **substantially reversible with folate/folinic acid**; some deficits (e.g., established retinopathy, neurodevelopmental sequelae, renal damage) may be irreversible.
- **Prognostic factors:** timeliness of molecular diagnosis and folate/folinic-acid initiation; severity of presenting phenotype (severe neonatal acidosis portends poor outcome); degree of residual MTHFD1 expression.
- No formal 5-/10-year survival statistics exist (case-level data only).

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12. Treatment

Core pharmacotherapy (metabolic/precision therapy). - **Folinic acid (5-formyltetrahydrofolate / leucovorin)** — preferred reduced folate that bypasses the dihydrofolate reductase step; enables immune reconstitution and hematologic correction

(P 27707659).

NCIT: **Leucovorin Calcium (C576)** / Folinic acid. - **Folic acid** — effective in milder/responsive patients; produced significant clinical improvement (P 42301236).

NCIT: **Folic Acid (C542)**. - **Hydroxocobalamin (vitamin B12)** — used to support remethylation; provided partial immune reconstitution with folate in the index case (P 23296427).

NCIT: **Hydroxocobalamin (C61805)**. - **Betaine** — remethylation agent to lower homocysteine (P 25633902).

NCIT: **Betaine (C61463)**. - **Methionine supplementation** — may be considered when hypomethioninemia present (inferred/supportive).

Supportive / adjunctive care. Anti-infective prophylaxis and immunoglobulin (IVIG) while immunodeficient; transfusion support for anemia; management of aHUS/TMA (supportive ± eculizumab per complement-mediated protocols, though MTHFD1-related TMA is metabolic in origin); treatment of autoimmune complications. Discontinuation of prophylaxis becomes possible after immune reconstitution on folinic acid (P 27707659).

Advanced/experimental therapeutics. No approved gene, cell, or RNA therapies. **Hematopoietic stem-cell transplantation is generally unnecessary** because the immunodeficiency is metabolically correctable — a key contrast with genetic SCIDs (this is an important treatment distinction). No registered disease-specific clinical trials identified.

Pharmacogenomics. Not established; genotype-guided care centers on choosing **reduced folate (folinic acid)** to bypass downstream steps.

Treatment strategy / algorithm. Suspect → urgent molecular diagnosis → immediate folinic/folic acid ± hydroxocobalamin ± betaine → supportive anti-infective/transfusion care → monitor hematology, homocysteine, and immune reconstitution → taper prophylaxis once reconstituted. **Early treatment is the strongest determinant of survival** (P 42301236).

13. Prevention

- **Primary prevention:** for the **Mendelian disease**, prevention is via **genetic counseling, carrier testing, and reproductive options** (prenatal/PGT) in at-risk families — not modifiable by lifestyle. For the **common R653Q modifier**, adequate **periconceptional folate** reduces associated NTD/CHD/pregnancy risk (GxE evidence; P 26408344).
- **Secondary prevention:** early recognition and prompt folate/folinic-acid therapy to prevent irreversible organ damage; cascade family screening.
- **Tertiary prevention:** anti-infective prophylaxis, IVIG, homocysteine control (betaine) to prevent vascular/renal complications until reconstitution.
- **Counseling:** autosomal recessive counseling — 25% recurrence risk for carrier couples.
- **Immunization / public health:** live vaccines contraindicated while immunodeficient; general folate fortification is a population-level protective measure for folate-pathway risks.

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *MTHFD1* is highly conserved. Mouse ortholog **Mthfd1** (NCBI Gene 108156; taxon *Mus musculus* NCBITaxon:10090). Orthologs exist across vertebrates and yeast (ADE3).
- **Natural disease:** no well-characterized spontaneous *MTHFD1*-deficiency disease reported in companion animals or wildlife (OMIA: none established).
- **Comparative biology:** the trifunctional C1-THF synthase and its one-carbon role are evolutionarily conserved from yeast to human; complete synthetase loss is **embryonic-lethal in mice**, underscoring conserved essentiality (P 23704330).
- **Zoonotic potential:** none (non-infectious genetic disease).

15. Model Organisms

- **Mouse (primary model).** A **Mthfd1 synthetase-specific hypomorph (Mthfd1S)** was engineered to model the R653Q variant by inactivating 10-formylTHF synthetase activity without disrupting protein expression or the other two activities (P 23704330).
- **Mthfd1S−/−:** embryonic lethal (~E10.5), developmentally delayed/abnormal — demonstrates essentiality.
- **Mthfd1S+/−:** reduced plasma/liver 10-formyl-THF, impaired de novo purine synthesis in MEFs, decreased neutrophil counts in pregnancy, increased embryonic developmental defects, and (on a separate cross) **increased congenital heart defects, chiefly ventricular septal defects** (P 23704330; 26408344).
- **Cellular / in vitro models:** patient-derived **fibroblasts** are the workhorse for enzymology and one-carbon flux studies (formate-incorporation assays, uracil-in-DNA measurement, Western blot of MTHFD1) (P 25548164; 32414565). **MEFs** used for purine-synthesis flux.
- **Phenotype recapitulation:** the mouse synthetase model recapitulates **developmental / purine-synthesis** aspects relevant to the *common R653Q variant* (NTD/CHD/pregnancy risk) but is a **partial** model of the human **biallelic Mendelian disease** (it does not reproduce the full megaloblastic-anemia + combined-immunodeficiency + aHUS syndrome). Patient fibroblasts best recapitulate the human dTMP/methionine flux defects.
- **Model limitations:** no reported mouse fully reproducing the human combined-immunodeficiency phenotype; ultra-rare human numbers limit genotype–phenotype modeling.
- **Resources:** MGI (Mthfd1), model described in Rozen-lab publications (P 23704330 P 26408344).

Supported and Refuted Hypotheses

Supported: - MTHFD1 deficiency is an AR loss-of-function disorder of the cytoplasmic/nuclear folate cycle (P 21813566; 32414565). - The hematologic + immune phenotype arises chiefly from **impaired nuclear de novo dTMP synthesis with uracil misincorporation**, with **purines relatively spared** (P 25548164). - The disorder is **folate/folinic-acid responsive**, and early treatment improves survival/immune reconstitution (P 27707659; 23296427; 42301236). - The common **R653Q** synthetase variant is a **distinct low-penetrance developmental risk modifier**, not the Mendelian disease (P 23704330; 26408344).

Refuted / clarified: - Not primarily a purine-synthesis disorder at the cellular level (purine flux preserved in patient cells) — refutes the intuitive "SCID via purine block" model as the main mechanism (P 25548164). - Not a mitochondrial folate defect — MTHFD1 is cytosolic/nuclear (distinct from MTHFD2/MTHFD1L). - HSCT is generally not required (metabolically correctable), distinguishing it from classical genetic SCID.

Limitations and Future Directions

- **Ultra-rare** (<~20 confirmed patients): frequencies, penetrance, and prognosis are estimated from case-level data; no registry/epidemiologic denominators.
- Mechanistic data on aHUS/neurologic branches remain partly **inferred** (homocysteine-mediated endothelial injury; hypomethylation) rather than directly demonstrated in patients.

- No mouse model of the **biallelic human disease**; no approved advanced therapeutics or trials.
- **Future needs:** a natural-history registry; systematic immunophenotyping and methylome profiling of patients; standardized treatment protocols (folinic acid dosing, betaine, B12); evaluation of newborn-screening biomarkers (homocysteine/methionine + macrocytosis).

Database provenance (verified during this investigation)

- **Disease/gene identifiers** (MONDO:0060611/CIMAH → OMIM:617780, Orphanet:658813, UMLS:C4540434, GARD:0026001, MedGen:1615364; MTHFD1 HGNC:7432, Ensembl ENSG00000100714, UniProt P11586, 14q23.3) — verified via **EBI OLS4/MONDO** and **MyGene.info**.
- **Common variant R653Q (rs2236225** = **GRCh38 14-64442127-G-A)** allele frequencies — verified via **gnomAD v4** (global genome AF 0.386; NFE 0.453 → ~20.5% QQ homozygotes).
- **Pathogenic variant spectrum** (64 P / 30 LP, predominantly splice/frameshift/nonsense; reference NM_005956.4) — verified via **NCBI ClinVar** (E-utilities).
- **Evidence tiers:** human clinical = case reports/series (PMIDs 21813566, 23296427, 25633902, 27707659, 32414565, 42301236); in vitro = patient fibroblast flux/enzymology (25548164, 32414565, 28265077); model organism = Mthfd1 synthetase mouse (23704330, 26408344); genetic-epidemiology = R653Q/NTD association studies (12384833, 16552426, 24977710, 22856873, 19130090).

Key References (PMID)

- 21813566 — First identification of MTHFD1 deficiency (exome).
- 23296427 — SCID from MTHFD1; response to B12+folate.
- 25548164 — Impaired nuclear de novo dTMP biosynthesis mechanism.
- 25633902 — Four new patients; folic/folinic acid treatment review.
- 27707659 — Precision diagnosis (incl. structural variant) and immune reconstitution.
- 32412981 — Review: immunodeficiency in inborn errors of B12/folate.
- 32414565 — Biochemical characterization (protein/activity) in patients.
- 42301236 — Two children (p.Arg173Cys); folate therapy and outcomes.
- 23704330 / 26408344 — Mthfd1 synthetase mouse models (R653Q); purine synthesis, NTD/CHD.

- 28265077 — Arsenic targets MTHFD1/SUMO-dependent nuclear dTMP synthesis.
- 12384833 — R653Q as maternal NTD risk factor (original, OR 1.52).
- 16552426 — Independent replication of maternal NTD risk (OR 1.49).
- 24977710 — Meta-analysis of MTHFD1 G1958A and NTD risk (9 studies).
- 22856873 — Candidate-gene screen confirming MTHFD1 R653Q among top NTD signals; NTD ~1/1000.
- 19130090 — MTHFD1 promoter variant (rs1076991) modifies NTD risk with R653Q.
